

MW 1.50 RSE Calculation Report – Cooling Tower Supporting Frame

Project: Erection of Supporting Structures for 9 Cooling Towers on Flat Roof

1. Introduction

Erection of supporting steel frame for 9 nos. air-cooled cooling towers (1500 kg each) on flat roof at 18 Kwai Hei Street, Kwai Chung. Complies with MW 1.50 Class I Type A (MWTGe.pdf Appendix VII 3.19). No modification to cantilevered slab.

2. Structural Synopsis

Hot-dip galvanized steel frame (UC 152×152×23 kg/m columns + SHS beams). Height < 1.5 m. Flat roof only.

3. Design Codes

MWTGe.pdf Appendix VII 3.19, CoP Structural Use of Steel 2011, Wind Effects 2019.

4. Material Strength

Grade S275, $p_y = 275 \text{ N/mm}^2$. Weld design strength = 220 MPa.

5. Loading (Step-by-Step)

DL = 15 kN/tower. LL per leg = 3.75 kN. Wind = 5.52 kPa (Z-dir governing). ULS: $1.4DL + 1.6LL + 1.2Wind_Z$.

6. Structural Analysis (STAAD.Pro)

Max: $F_x = 39.1 \text{ kN}$, $M_{xx} = 16.1 \text{ kNm}$, $V_y = 38.4 \text{ kN}$ (ULS D+L+Wz).

7. Member Capacity

$M_{cap} = 62.44 \text{ kNm} > 16.1 \text{ kNm}$ O.K. $V_{cap} = 380.2 \text{ kN} > 38.4 \text{ kN}$ O.K. Interaction = $0.12 < 1.0$ O.K.

8. Deflection Check

$\delta_{max} = 3.42 \text{ mm} < L/200 = 4.25 \text{ mm}$ O.K.

9. Anchor & Weld

Hilti HST3-R M12: 92% util O.K. 5mm fillet weld: $174.3 \text{ MPa} < 220 \text{ MPa}$ O.K.

10. Conclusion + RSE Sign

All members adequate. Structure safe.

RSE Signature: _____ Chan Pak Ming RSE 95/13